

Mekelle Institute Of Technology

Software Engineering Individual Assignment

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1. What are the key differences between functional and non-functional requirements?
    - Functional requirements describe the specific features and operations the system must perform.
    - Non-functional requirements define the quality attributes such as performance, security, and usability. In short, functional is *what the system does*, while non-functional is *how well it does it*.
2. How do you distinguish between user requirements and system requirements?
    - User requirements are written in simple language and describe what the users expect from the system at a high level.
    - System requirements are more detailed and technical, providing precise descriptions for developers to implement. They act as a bridge between business needs and the actual solution.
3. What are the main activities involved in the Requirements Engineering process?
    - Requirements engineering involves elicitation, analysis, specification, validation, and management. First, requirements are gathered from stakeholders and then analyzed for clarity and feasibility. Finally, they are documented, checked for correctness, and controlled as changes occur.
4. What is the role of a software analyst in gathering requirements?
    - A software analyst communicates with stakeholders to understand their needs and business processes. They convert these needs into clear, structured, and testable requirements. They also help resolve conflicts and ensure the requirements are complete and consistent.
5. How do you manage changing requirements during development?
    - Changing requirements are handled through a formal change management process. Each change is analyzed for its impact on cost, time, and system design before approval. Using iterative methods like Agile also makes it easier to adapt to changes.

6. What is a Software Requirements Specification (SRS) document?
    - A Software Requirements Specification is a formal document that describes all functional and non-functional requirements of a system. It serves as an agreement between stakeholders and developers. It also acts as a reference for design, development, and testing.
7. What is the purpose of system modeling in software engineering?
    - System modeling is used to create visual representations of the system. It helps stakeholders and developers understand the structure and behavior before implementation begins. This reduces ambiguity and improves communication.
8. Explain the difference between behavioral modeling and structural modeling.
    - Behavioral modeling focuses on how the system behaves and interacts over time. Structural modeling shows the static organization of the system and its components. Together, they provide both dynamic and static views of the system.
9. What is a Use Case diagram and what components does it include?
    - A Use Case diagram shows how external users, called actors, interact with the system. It identifies the main system functions through use cases. It also includes relationships such as association, include, and extend.
10. Describe the purpose of Entity Relationship Diagrams (ERD) in modeling.
    - An Entity Relationship Diagram is used to model the logical structure of a database. It shows entities, their attributes, and the relationships between them. This helps in designing a well-organized and consistent data system.
11. What are the key principles of software design?
    - Software design should follow principles such as modularity, abstraction, and separation of concerns. High cohesion and low coupling are important for maintainability and flexibility. Simplicity and reusability also improve the overall quality of the system.

12. What is architectural design and how does it differ from detailed design?

- Architectural design defines the overall structure of the system and how major components interact. Detailed design focuses on the internal logic and algorithms of each component. It provides the information needed for actual coding.

13. What is a user interface (UI) design process?

- The UI design process starts with understanding user needs and system requirements. Designers then create wireframes and prototypes to represent the interface. These designs are tested and refined based on user feedback.

14. What is a Data Flow Diagram (DFD) and what does it represent?

- A Data Flow Diagram shows how data moves through a system from input to output. It identifies processes, data stores, external entities, and data flows. It helps in understanding the functional flow of information within the system.